29/01/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

Regd. Office: Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 7

Answer Key

1.	(3)	37.	(2)	73.	(4)	109.	(4)	145.	(1)
2.	(2)	38.	(4)	74.	(3)	110.	(2)	146	(4)
3.	(4)	39.	(4)	75.	(4)	111.	(1)	147.	(4)
4.	(1)	40.	(1)	76.	(4)	112.	(3)	148.	(2)
5.	(1)	41.	(1)	77.	(3)	113.	(4)	149.	(1)
6.	(2)	42.	(3)	78.	(3)	114.	(2)	150.	(3)
7.	(3)	43.	(2)	79.	(2)	115.	(4)	151.	(2)
8.	(3)	44.	(2)	80.	(2)	116.	(3)	152.	(2)
9.	(3)	45.	(3)	81.	(3)	117.	(2)	153.	(4)
10.	(2)	46.	(2)	82.	(3)	118.	(3)	154.	(2)
11.	(1)	47.	(2)	83.	(3)	119.	(2)	155.	(2)
12.	(4)	48.	(3)	84.	(1)	120.	(3)	156.	(2)
13.	(1)	49.	(4)	85.	(2)	121.	(4)	157.	(2)
14.	(3)	50.	(1)	86.	(2)	122.	(4)	158.	(4)
15.	(4)	51.	(3)	87.	(2)	123.	(1)	159.	(3)
16.	(3)	52.	(4)	88.	(3)	124.	(2)	160.	(4)
17.	(4)	53.	(2)	89.	(3)	125.	(2)	161.	(1)
18.	(2)	54.	(1)	90.	(1)	126.	(3)	162.	(2)
19.	(2)	55.	(1)	91.	(4)	127.	(1)	163.	(4)
20.	(1)	56.	(2)	92.	(1)	128.	(2)	164.	(1)
21.	(4)	57.	(2)	93.	(2)	129.	(3)	165.	(4)
22.	(2)	58.	(2)	94.	(2)	130.	(4)	166.	(3)
23.	(2)	59.	(1)	95.	(3)	131.	(4)	167.	(3)
24.	(3)	60.	(2)	96.	(4)	132.	(1)	168.	(1)
25.	(4)	61.	(1)	97.	(2)	133.	(4)	169.	(2)
26.	(1)	62.	(2)	98.	(3)	134.	(3)	170.	(1)
27.	(1)	63.	(2)	99.	(4)	135.	(3)	171.	(1)
28.	(2)	64.	(1)	100.	(4)	136.	(1)	172.	(4)
29.	(1)	65.	(2)	101.	(1)	137.	(4)	173.	(4)
30.	(2)	66.	(3)	102.	(1)	138.	(2)	174.	(3)
31.	(2)	67.	(4)	103.	(2)	139.	(3)	175.	(1)
32.	(3)	68.	(1)	104.	(3)	140.	(4)	176.	(2)
33.	(3)	69.	(4)	105.	(3)	141.	(3)	177.	(4)
34.	(2)	70.	(2)	106.	(1)	142.	(4)	178.	(4)
35.	(3)	71.	(3)	107.	(4)	143.	(2)	179.	(4)
36.	(1)	72.	(4)	108.	(2)	144.	(4)	180.	(3)

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3:00 Hrs.

Test - 7

Hints and Solutions

PHYSICS

1. Answer (3)

$$I = \frac{E}{\frac{R}{n} + R}$$

$$\frac{I}{9} = \frac{E}{nR + R}$$

$$9 = \frac{n+1}{\left(\frac{1}{n}+1\right)}$$

$9 = n$

2. Answer (2)

$$R = \rho\frac{\ell}{A}$$

$$\ell A = 3\ell A_1$$

$$A_1 = \left(\frac{A}{3}\right)$$

$$R_1 = \rho\frac{3\ell}{\frac{A}{3}} = 9R = 9R$$

3. Answer (4)

$V_P - V_Q = +1 \times 2 - 6 + 1 \times 2 + 2$

$V_P - V_Q = 2 - 6 + 2 + 2 = 0$

4. Answer (1)

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{\frac{\ell}{2\sigma_{eff}A}} = \frac{1}{\frac{\ell}{\sigma A}} + \frac{1}{\frac{\ell}{2\sigma A}}$$

$$2\sigma_{eff} = \sigma + 2\sigma$$

$$\sigma_{eff} = \frac{3\sigma}{2}$$

5. Answer (1)

Let the resistance of the shorter part *MN* be X. Total resistance is 20 Ω. Hence, the resistance of longer *MN* part will be (20 – *x*).

$$R_{eq} = \frac{(20-x)x}{20-x+x} = 1.8$$

$x = 2\ \Omega$

The resistance per unit length 1 Ω/m so, length of the shorter part = 2 m.

6. Answer (2)

$V < E$

$E = V + Ir$

For first case $E = 12 + \frac{12}{16}r$ … (I)

For second case $E = 11 + \frac{11}{10}r$ … (II)

On solving (I) and (II)

$$r = \frac{20}{7}\Omega$$

7. Answer (3)

$$\frac{40}{20} = \frac{R}{80}$$

$R = 160\ \Omega$

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8. Answer (3)

$$r = R\left(\frac{\ell_1}{\ell_2} - 1\right)$$

$$r = 6\left(\frac{70}{60} - 1\right) \Rightarrow r = 1\ \Omega$$

9. Answer (3)

$V = IR$

$= I\rho\frac{\ell}{A}$

$V = J\ \rho\ell$

$$\rho = \frac{V}{J\ell} \Rightarrow \rho = \frac{115}{1.4\times10^4\times10} = 8.2\times10^{-4}\Omega m$$

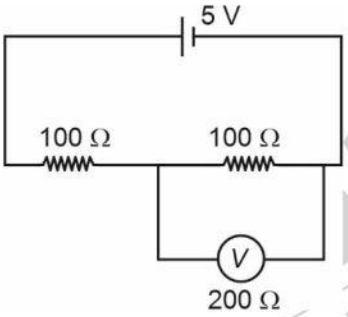
10. Answer (2)

$$P = \frac{V^2}{R}$$

$$P_1 = \frac{V^2}{\left(\frac{R}{2}\right)} = 2\frac{V^2}{R}$$

$= 2 \times 60 = 120$ watt

11. Answer (1)



$$K_{eff} = \frac{200\times100}{200+100} + 100$$

$$R_{eff} = \frac{200}{3} + 100 = \left(\frac{500}{3}\right)$$

$$I = \frac{V}{R_{eff}}$$

$$I = \frac{5}{\frac{500}{3}} = \left(\frac{3}{100}\right)$$

Reading of voltmeter will be

$$= \frac{3}{100}\times\frac{200}{3} = 2\ V$$

12. Answer (4)

$$I_{2\Omega} = \frac{4\times15}{15+6} = \frac{4\times15}{21} = \frac{20}{7}$$

13. Answer (1)

$V = IR$

$R = R\ (1 + \alpha T)$

$R_1 > R_2$

$T_1 > T_2$

14. Answer (3)

$V_0 - V_A = 2 \times 4$

$V_0 - V_B = 2 \times 6$

$V_A - V_B = 4$ V

15. Answer (4)

$$I = \frac{6}{5+10} = 0.4\ A$$

$$2.4 = 0.4\times\frac{10}{100}L$$

$L = 60$ cm

16. Answer (3)

$2x + 3x = 100$

$5x = 100$

$x = 20$ cm

$2x = 20 \times 2 = 40$ cm

17. Answer (4)

$$\frac{V^2}{R_{eff}} = P$$

$$\frac{(20)^2}{\frac{10R}{10+R}} = 50$$

$400\ (10 + R) = 500R$

$40 + 4R = 5R \Rightarrow R = 40\ \Omega$

18. Answer (2)

Kirchhoff's loop law follow the law of conservation of energy and Kirchhoff' junction law follow the law of conservation of charge.

19. Answer (2)

$$\varepsilon_{eff} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$= \frac{2\times2 + 3\times1}{1+2} = \frac{7}{3}\ V$$

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20. Answer (1)

$i = \frac{dQ}{dt}$ is uniform.

21. Answer (4)

On increasing the temperature of conductor its resistance increases because the relaxation time decreases.

22. Answer (2)

$I = \frac{V}{R_{eff}}$

$I = \frac{4}{5} = 0.8$ A

23. Answer (2)

$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

$F_1 = F_2 = \frac{\mu_0 I^2}{2\pi d}$

$F = \frac{\sqrt{3}\,\mu_0 I^2}{2\pi d}$

24. Answer (3)

$B = \frac{\mu_0 I}{2R}$

$2\pi R = n2\pi r_1$

$r_1 = \frac{R}{n}$

$B_1 = \frac{n\mu_0 I}{2r_1}$

$B_1 = \frac{n\mu_0 I}{2\left(\frac{R}{n}\right)}$

$B_1 = \frac{n^2\mu_0 I}{2R}$

$B_1 = n^2 B$

$= 4^2 B = 16B$

25. Answer (4)

When a charged particle enters in a uniform magnetic field its speed remains same therefore kinetic energy remains constant.

26. Answer (1)

$B = \frac{\mu_0 I}{2R} \times \frac{\phi}{2\pi}$

27. Answer (1)

$B = \frac{2\mu_0 I}{2\pi d} + \frac{3\mu_0 I}{2\pi d}$

$B = \frac{5\mu_0 I}{2\pi d}$

$B_1 = \frac{3\mu_0 I}{2\pi d}$

$\frac{B}{B_1} = \frac{5}{3} \Rightarrow B_1 = \left(\frac{3B}{5}\right)$

28. Answer (2)

$v = \frac{E}{B}$

$v = \frac{1200}{0.2}$

$v = 6.0 \times 10^3$ m/s

29. Answer (1)

$B_{in} = 0$

$B_{out} = \frac{\mu_0 i}{2\pi d}$

30. Answer (2)

$B = B_1 + B_2$

$B_1 = \frac{\mu_0 I}{4R}$

$B_2 = \frac{\mu_0 I}{4\pi R}$

$B = \frac{\mu_0 I}{4R}\left(\frac{1}{\pi} + 1\right)$

31. Answer (2)

$\vec{F} = q(\vec{v} \times \vec{B})$

$= -4 \times 10^{-6}\left[\left\{(3\hat{i} + 4\hat{j}) \times 10^6\right\} \times 1\hat{j}\right]$

$= -4(3\hat{i} \times \hat{j} + 4\hat{j} \times \hat{j})$

$= -12\hat{k}$ N

32. Answer (3)

$M = NIA$ | $A = \pi\left(\frac{2}{\pi}\right)^2$

$2\pi R = 4$ | $= \frac{4}{\pi}$

$R = \frac{2}{\pi}$ | $M = \frac{8}{\pi}$ Am^2

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33. Answer (3)

$M = NIA$

$M = NI\pi r^2$

$M \propto r^2$

34. Answer (2)

$R = \frac{Mv}{qB}$

$R \propto v$

$\frac{R_1}{R_2} = \frac{4}{3}$

35. Answer (3)

$\vec{F} = i\left(\vec{\ell} \times \vec{B}\right)$

$|\vec{F}| = 5 \times 4 \times 10^{-2} \times 0.4$

$= 8 \times 10^{-2}$ N

36. Answer (1)

$r = \frac{mv}{qB} \Rightarrow r \propto v$

37. Answer (2)

$V = I_g R_g + I_g R$

$4 = (16 + R)\, 2 \times 10^{-3}$

$R + 16 = 2000 = 1984\ \Omega$

38. Answer (4)

Current sensitivity $\frac{\phi}{I} = \frac{NAB}{K}$

Voltage sensitivity = $\frac{\phi}{V} = \left(\frac{NAB}{K}\right)\frac{I}{V} = \frac{NAB}{K} \times \frac{1}{R}$

Voltage sensitivity

$$= \text{Current sensitivity} \times \frac{1}{\text{Resistance}}$$

When current sensitivity doubled then resistance in the circuit will also double, hence voltage sensitivity remains same.

39. Answer (4)

$\vec{\tau} = \vec{M} \times \vec{B}$

$M = NIA$

40. Answer (1)

$P = \vec{F} \cdot \vec{v}$ and $F \perp v$ and $F \perp B$

41. Answer (1)

$\mu = \frac{qL}{2M}$

$L = \frac{MR^2\omega}{2}$

$\mu = \frac{qR^2\omega}{4}$

42. Answer (3)

$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$

$B = \frac{\mu_0 I a^2}{2(3a^2 + a^2)^{3/2}}$

$B = \frac{\mu_0 I}{16a}$

43. Answer (2)

Ampere's circuital law can be derived from Biot-Savart's law.

44. Answer (2)

$P = \frac{2\pi m(v\cos\theta)}{qB}$

$r = \frac{mv\sin\theta}{qB}$

$\frac{P}{r} = \frac{8\pi}{3} \Rightarrow r = \frac{3P}{8\pi}$

45. Answer (3)

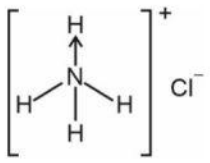
$B_{end} = \frac{\mu_0 nI}{2}$

CHEMISTRY

46. Answer (2)

Higher is electronegativity and oxidation state of central atom greater will be acidic strength.

47. Answer (2)



It has covalent, ionic and coordinate bond.

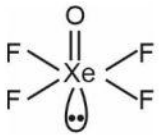
48. Answer (3)

$2NO + N_2O_4 \xrightarrow{250\ K} \underset{\text{Blue solid, acidic}}{2N_2O_3}$

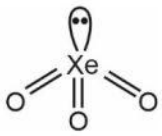
49. Answer (4)

From HF to HI; acidic strength increases because bond strength decreases.

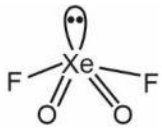
50. Answer (1)



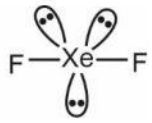
sp^3d^2; square pyramidal

XeO_3; 

sp^3; pyramidal

XeO_2F_2; 

sp^3d; see-saw

XeF_2; 

sp^3d; Linear

51. Answer (3)

Hair cream is an example of emulsion.

52. Answer (4)

Chemical adsorption increases with the increase of temperature.

53. Answer (2)

Nickel is purified by Mond process.

54. Answer (1)

$FeO + CO \xrightarrow{T < 1073\,K} Fe + CO_2$

55. Answer (1)

Calamine is an ore of zinc.

56. Answer (2)

$r = k(CH_3COCH_3)^{3/2}[O_2]^{1/2}$

$\therefore$ Order of reaction is $\frac{3}{2}+\frac{1}{2}=2$

Unit of k is = $Mol^{-1}\ L\ s^{-1}$

$\Rightarrow$ Unit of rate of reaction is $Mol\ L^{-1}\ s^{-1}$

57. Answer (2)

Acid hydrolysis of ethyl acetate is a pseudo first order reaction.

$\therefore\ r = k[CH_3COOC_2H_5]^1[H_2O]^0$

58. Answer (2)

Slow step of mechanism decides the rate expression.

$\therefore\ r = k[NO_2][F_2]$

The overall order of the reaction will be 2.

59. Answer (1)

The thermal decomposition of HI on gold surface is an example of zero order reaction.

60. Answer (2)

For first order

$\therefore\ k = \frac{2.303}{t}\log\frac{[R]_0}{[R]_t}$

$k = \frac{2.303}{15}\log\frac{100}{12.5}$

$k = \frac{2.303}{15}\log 8$

$k = \frac{2.303}{15}\times 3\log 2$

$\Rightarrow\ t_{1/2} = \frac{0.693}{k}$

$t_{1/2} = \frac{0.693\times 15}{2.303\times 3\log 2} = 5\ \text{minute}$

61. Answer (1)

Freundlich adsorption isotherm may be mathematically as $\frac{x}{m} = k\cdot p^{1/n}$ where n > 1.

62. Answer (2)

For spontaneous adsorption

ΔH = –ve, ΔS = –ve, ΔG = –ve

63. Answer (2)

Higher is the valency of the active ion, the greater will be its power to precipitate the sol.

$\therefore$ The reciprocal of coagulation value is regarded as the coagulating power.

64. Answer (1)

Pepsin converts proteins into amino acids in stomach.

65. Answer (2)

Temperature coefficient is 2.

$\therefore\ r = 2^6 = 64$

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66. Answer (3)

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 \times T_2}\right)$$

$\Rightarrow \quad k_2 = 3k_1$

$T_2 = 40°C = 313\,K$

$T_1 = 30°C = 303\,K$

$$\therefore \quad \log\frac{3k_1}{k_1} = \frac{E_a}{2.303 \times 8.314}\left(\frac{313 - 303\,K}{313 \times 303\,K}\right)$$

$$\Rightarrow \quad \log 3 = \frac{E_a}{2.303 \times 8.314}\left(\frac{10}{94839}\right)$$

$$\Rightarrow \quad E_a = 0.477 \times \frac{2.303 \times 8.314 \times 94839}{10}$$

$= 86618.2\ J\ mol^{-1}$

$E_a = 86.62\ kJ\ mol^{-1}$

67. Answer (4)

For zero order; $t_{1/2} = \frac{[R]_0}{2k}$

$\therefore$ $t_{1/2}$ for zero order reaction is directly proportional to the initial concentration of the reactants.

68. Answer (1)

Arrhenius proposed the following empirical equation relating the rate constant k and temperature T (in Kelvin), $k = A.\,e^{-E_a/RT}$

69. Answer (4)

$$A + 3B \rightarrow 2C + \frac{1}{2}D$$

$$\text{Rate} = -\frac{d[A]}{dt} = -\frac{d[B]}{3dt} = \frac{1}{2}\frac{d[C]}{dt} = \frac{2d[D]}{dt}$$

70. Answer (2)

$r_1 = k[C]^x[D]^y \ldots (I)$

$r_2 = k[3C]^x[D]^y \ldots (II)$

$r_3 = k[3C]^x[3D]^y \ldots (III)$

$II \div I$

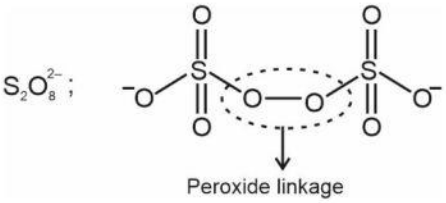
$\frac{3r_1}{r_1} = 3^x;\ x = 1$

$III \div II$

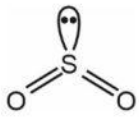
$\frac{9r_1}{3r_1} = 3^y;\ y = 1$

rate = [C] [D]

71. Answer (3)



72. Answer (4)



$\Rightarrow$ Hybridisation = sp^2

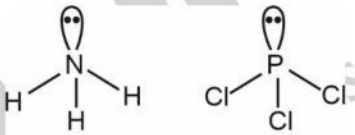
Shape = angular

73. Answer (4)

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

74. Answer (3)

NH_3 is isostructural with PCl_3



75. Answer (4)

The correct order of increasing boiling point is

$PH_3 < AsH_3 < NH_3 < SbH_3$

76. Answer (4)

Metallic oxides are positively charged sols.

e.g. TiO_2 sol.

77. Answer (3)

The catalytic process in which the reactants and the catalyst are in different phase is known as heterogeneous catalysis.

78. Answer (3)

For soaps, the CMC is 10^{-4} to 10^{-3} mol L^{-1}.

79. Answer (2)

$2FeCl_3(aq) + 6NH_4OH(aq) \rightarrow$

$Fe_2O_3.xH_2O(s) + 6NH_4Cl(aq)$

(Brown ppt)

80. Answer (2)

$$\underset{\text{White phosphorus}}{\overset{0}{P_4}} + 3NaOH + 3H_2O \rightarrow \underset{\text{Phosphine}}{\overset{-3}{PH_3}} + \underset{\text{Sodium Hypophosphite}}{3NaH_2\overset{+1}{P}O_2}$$

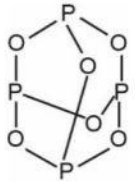
81. Answer (3)

$$I_2 + 10HNO_3 \rightarrow \underset{\text{Iodic acid}}{2HIO_3} + 10NO_2 + 4H_2O$$

82. Answer (3)

$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

83. Answer (3)



P_4O_6 (It has 6 bridging oxygen atoms.)

84. Answer (1)

Correct order of acidic strength is

$HClO < HClO_2 < HClO_3$

85. Answer (2)

F_2 molecule has low bond dissociation enthalpy than Br_2 and Cl_2 because relatively large electron-electron repulsion among lone pairs in F_2 molecules.

86. Answer (2)

$\underset{x}{HNO_3}$; $+1 + x - 6 = 0$

$x = +5$

$\underset{x}{N_3H}$; $\Rightarrow 3x + 1 = 0$

$x = -\frac{1}{3}$

$\underset{x}{N_3H}$; $\Rightarrow x + 3 = 0$

$x = -3$

87. Answer (2)

Chromatographic method is based on the principle of adsorption.

88. Answer (3)

Copper matte contains Cu_2S + FeS

89. Answer (3)

Wrought iron is the purest form of iron.

90. Answer (1)

Bronze $\rightarrow$ An alloy of Cu and Sn

BOTANY

91. Answer (4)

Reproductive meristem grows to form inflorescence axis over which floral primordia are formed, then primordia grow into floral buds.

92. Answer (1)

Proximal end of filament is attached to thalamus or petal of the flower.

93. Answer (2)

Anther is tetragonal in transverse section.

94. Answer (2)

Breeding, marketing, cultivation etc. of flowers is floriculture.

95. Answer (3)

Production of microspores/pollen grains from microspore mother cell by meiosis is microsporogenesis.

96. Answer (4)

Tapetum is nutritive and secretory in function.

97. Answer (2)

Size of pollen grains is 25-50 micrometer in diameter

Pollen architecture also includes its size and shape.

Pollens are fossilised due to sporopollenin.

98. Answer (3)

Middle layer of anther wall is 1–3 layers.

99. Answer (4)

Microspore $\xrightarrow{\text{Mitosis}}$ vegetative cell and generative cell.

100. Answer (4)

60% angiosperms show shedding of pollen grains at two celled stage. Generative cell divides into two male gametes in pollen tube.

101. Answer (1)

No enzyme is so far known to degrade sporopollenin.

102. Answer (1)

Basal parts of pistils are fused and styles and stigma are free in *Hibiscus*.

103. Answer (2)

Chalaza is the basal part of ovule from where integuments arise.

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104. Answer (3)

$$\text{Nucellus}(2n) \rightarrow \underset{(2n)}{\text{MMC}} \xrightarrow{R/D} \underset{(n)}{\text{Megaspores}}$$

105. Answer (3)

The synergids have special cellular thickenings towards micropylar end called filiform apparatus.

106. Answer (1)

Colourful flower with nectar is the attraction for pollinators like insects, birds or other animals.

107. Answer (4)

Pollen-pistil recognisation is followed by promotion or inhibition of the pollen germination.

108. Answer (2)

The zygote gives rise to the pro-embryo and subsequently to the other forms as in option (2).

109. Answer (4)

Occasionally, in some seeds such as beet and black pepper, remnant of nucellus are also present, called perisperm.

110. Answer (2)

Endosperm development occurs before embryo development.

111. Answer (1)

Strawberry is false fruit.

112. Answer (3)

A *Ficus* tree produces billions of seeds.

113. Answer (4)

Some species of Asteraceae and grass family have evolved apomixis.

114. Answer (2)

Number of chromosomes in each cell of

Endosperm = 3n = 3 × 8 = 24

Egg apparatus = n = 8

Nucellus = 2n = 2 × 8 = 16

Leaf cell = 2n = 16

115. Answer (4)

Human knew one of the cause of variation was hidden in sexual reproduction as early as 8000-1000 B.C.

116. Answer (3)

Punnett square was developed by a British geneticist R.C. Punnett.

117. Answer (2)

$I^A i^o \times I^B i^o$

AB B A O

Probability of blood group 'A' = ¼ = 25%.

118. Answer (3)

Human skin colour is polygenic trait and was studied by Davenport.

119. Answer (2)

Myotonic dystrophy is autosomal dominant disorder.

120. Answer (3)

$$TtYy \times ttYy$$
$$\downarrow$$
$$\text{Phenotypic ratio} = \frac{3}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}$$

121. Answer (4)

All the daughters will be affected if their father has X-linked dominant trait.

122. Answer (4)

Change in a subset of chromosome number is called aneuploidy. Down's syndrome is example of 2n + 1.

123. Answer (1)

A person having Klinefelter's anomaly has 47 chromosomes instead of 46. XXY- abnormality is seen in Karyotype.

124. Answer (2)

If two genes are completely linked, the phenotypic ratio is 3 : 1 instead of 9 : 3 : 3 : 1.

125. Answer (2)

Drosophila completes its life cycle in about two weeks.

126. Answer (3)

Birds : ZW – ZZ type

Human : XX-XY type

127. Answer (1)

HNO_2, Proflavin, Acridine dye, UV-radiation,

Gamma radiation, X-rays etc. are mutagens.

128. Answer (2)

Queen Victoria was a carrier of haemophilia.

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129. Answer (3)

3 : 1 (dominance) × 1 : 2 : 1 (Incomplete dominance)

= 3 : 6 : 3 : 1 : 2 : 1

130. Answer (4)

Mendel performed hybridization experiments on pea plant from 1856 to 1863.

131. Answer (4)

Sickle-cell anaemia is due to replacement of glutamic acid at 6th position of the β-chain of haemoglobin by valine.

132. Answer (1)

Chronic myelogenous leukemia is one of the example of cancer, caused by chromosomal aberrations.

133. Answer (4)

$$\frac{n}{2}(n+1)=\frac{5}{2}\times 6=15=\text{genotypes}$$

134. Answer (3)

1% of recombinants means the distance between genes is 1 cM.

135. Answer (3)

Types of gametes = 2^n,

n = Number of heterozygous pair = $2^3 = 8$

ZOOLOGY

136. Answer (1)

Scrotum provides 2 to 2.5°C temperature lower than abdominal cavity which is required for proper spermatogenesis.

137. Answer (4)

FSH receptors are located on surface of Sertoli cells which secrete certain factors for spermiogenesis

138. Answer (2)

Urethra is common tract for both urine and semen in humans and other mammals only in males. In females, urethra is only part of excretory tract.

139. Answer (3)

During pregnancy, hormones such as thyroxine, cortisol, prolactin, oestrogen and progesterone increases to support foetal growth, metabolic changes in the mother and maintenance of pregnancy.

140. Answer (4)

Androgens are required for development of seminiferous tubules at puberty and both androgens and FSH are required for spermatogenesis.

141. Answer (3)

Oviducts/fallopian tubes are site for fertilization in human.

142. Answer (4)

LH reaches at its peak level just before ovulation called LH surge. As a result it comes out in urine.

143. Answer (2)

Insemination is the transfer of sperms into the female genital tract.

144. Answer (4)

Hypothalamus starts secreting hormone GnRH in pulsatile manner at a certain age (puberty) responsible for changes during puberty by acting through pituitary gland and gonads. So, GnRH is switch on hormone for puberty.

145. Answer (1)

Vasa efferentia transmit sperms from testis to epididymis.

146 Answer (4)

All these statements are true for a reproductively healthy society.

147. Answer (4)

A women plays a key role in the continuity of the family and human species.

148. Answer (2)

Amniotic fluid present in amniotic cavity around foetus is used for diagnosis of prenatal genetic disorders.

149. Answer (1)

Productivity is ability of organisms to produce their offsprings. Inability in males to copulate due to failure in erection of penis is called impotency.

150. Answer (3)

IVF refers to "*in vitro*" fertilization in which fertilization is done outside the body.

151. Answer (2)

FSH acts on Sertoli cells to secrete certain factors which help in spermiogenesis.

152. Answer (2)

Spirally arranged mitochondria are present in middle piece of sperm around axial filament.

153. Answer (4)

Each sperm contains either X or Y-chromosome but in eggs produced by females, only one X-chromosome is present in all ova along with autosomes.

154. Answer (2)

A	Acrosome	Contain enzymes which help in fertilization
B	Middle piece	Contains mitochondria which produces energy for movement of tail
C	Tail	Facilitates sperm motility

155. Answer (2)

In human females, oogenesis starts during foetal life. Meiosis-I starts in primary oocyte but arrest at diplotone stage of prophase-I.

156. Answer (2)

Bulbovestibular gland is accessory gland in females. Seminal vesicles and bulbourethral glands are paired accessory glands in males.

157. Answer (2)

Parturition – Delivery of mature foetus

Gestation – Period between fertilization and birth

Ovulation – Release of ovum from ovary

Implantation – Attachment of embryo to endometrium of uterus

158. Answer (4)

Spermatids are haploid cells whereas other given cells are diploid cells of testis or ovary.

159. Answer (3)

Inner cell mass is responsible for formation of embryo. So, also called embryoblast.

160. Answer (4)

Meiosis-II starts within secondary oocyte and completed during fertilization after entry of sperm within secondary oocyte.

161. Answer (1)

In GIFT, ovum is transferred in fallopian tube of female who can not produce ovum but provides a suitable environment for *in-vivo* fertilization and implantation.

162. Answer (2)

IVF involves the fertilization outside the body followed by embryo transfer within uterus or fallopian tube.

163. Answer (4)

Genital herpes, Hepatitis-B and AIDS are incurable STIs.

164. Answer (1)

Trichomoniasis is caused by infection of a protozoan named *Trichomonas vaginalis*

165. Answer (4)

Multiload-375 is a copper releasing IUD.

166. Answer (3)

Leydig cells secrete androgens in interstitium of testis. Corpus luteum persists after ovulation under effect of LH and during pregnancy under effect of hCG.

167. Answer (3)

Number of days/year = 365-366

So number of eggs produced per year $= \frac{365}{28} \approx 13$

168. Answer (1)

Placenta is structural and functional unit between foetus and maternal body (uterus).

169. Answer (2)

Infundibulum lead to a wider part of oviduct called ampulla.

170. Answer (1)

Male produces sperms with either X or Y chromosomes and are responsible for sex determination.

171. Answer (1)

Oxytocin is synthesized during pregnancy and lactation.

172. Answer (4)

GnRH is switch on hormone for menarche. Cycle of events starting from one menstruation till the next one is called the menstrual cycle. Cycle starts with menstrual phase and ends will luteal phase.

173. Answer (4)

Prolactin is a lactogenic hormone and is not involved in parturition.

174. Answer (3)

(a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

175. Answer (1)

Foetal ejection reflex triggers release of oxytocin form maternal pituitary gland which causes strong uterine contraction by positive feedback mechanism for parturition.

176. Answer (2)

Periodic abstinence, coitus interruptus and lactational amenorrhoea are natural methods of contraception.

177. Answer (4)

Increase in number of people in reproducible age is responsible for population explosion in India.

178. Answer (4)

An ideal contraceptive is effective and reversible. Sterilization is not an ideal contraceptive due to its poor reversibility although it is most effective.

179. Answer (4)

Only estrogens are not used as emergency contraceptive as it has more side effects.

180. Answer (3)

Parturition is common (same) for both test tube baby and normal baby.

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